

scripts/ver_linux in kernel tree

When you write a shell program, sometimes it is necessary to get information about the system in which it is running (especially the installed components and versions).

For example, if you want to check the GCC version, you may run the command shown below, with the result as shown:

```
gcc -dumpversion 2>&1 | awk \
'NR==1{print "Gnu C", $1}'
Gnu C      4.4.3
```

There are several such checks that are performed by the scripts/ver_linux shell script in the kernel source tree. The screen shot in Figure 3 shows the results of executing this script.

```
File Edit View Terminal Help
Compare to the current minimal requirements in Documentation/Changes.

Linux GNU-BOX 2.6.32-21-generic #32-Ubuntu SMP Fri Apr 16 08:10:02 UTC 2010 i686 GNU/Linux

Gnu C      4.4.3
Gnu make   3.81
binutils   2.20.1
util-linux 2.17.2
mount      support
module-init-tools 3.11.1
elfutils   1.41.11
pcmciautils 014
Linux C Library 2.11.1
Dynamic linker (ldd) 2.11.1
Procs      3.2.8
Net-tools  1.60
Kbd        1.15
Sh-utils   7.4
Wireless-tools 30
Modules loaded 30
vboxvideo  drm agpgart binfmt_misc vboxsf snd_intel8x0 snd_ac97_codec ac97_bus snd_pcm_oss snd_mixer_oss snd_pcm snd_seq_dummy snd_seq_oss snd_seq_midi snd_rawmidi snd_seq_midi_event snd_seq_pdrv snd_timer snd_seq_device fbcon tileblit font bitblit softcursor parport pc_joystick snd_psmouse serio r
```

Figure 3: Getting system-specific details

this, and it tells the system to wait till data arrives at `sk_receive_queue`:

```
int sk_wait_data(struct sock *sk, long *timeo)
{
    int rc;
    DEFINE_WAIT(wait);

    prepare_to_wait(sk->sk_sleep, &wait, TASK_INTERRUPTIBLE);
    set_bit(SOCK_ASYNC_WAITDATA, &sk->sk_socket->flags);
    rc = sk_wait_event(sk, timeo, !sk->sk_receive_queue);
    empty(sk->sk_receive_queue);
    clear_bit(SOCK_ASYNC_WAITDATA, &sk->sk_socket->flags);

    finish_wait(sk->sk_sleep, &wait);
    return rc;
}
EXPORT_SYMBOL(sk_wait_data);
```

Here `sk` represents the socket to wait on, and `timeo` is used to set the time till when it should wait. You may see that we are not performing any checks; this is due to the fact that the state of the socket can only be changed under lock.

We have seen some of the important functions that perform the initialisation and basic operations. We will discuss other relevant functions in the next edition.

Happy kernel hacking! 

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